

UNIVERSAL FIELDS

Quantum field theory (QFT) is a broad framework that treats all particles as ripples in their underlying quantum fields. For instance, an electron emerges when the electron field is excited beyond a certain limit. Due to the uncertainty principle (see pp.42–43), these fields constantly froth with particles and antiparticles appearing from nothingness and vanishing an instant later. This theoretical framework incorporates theories such as the Standard Model (see pp.124–25).

QUANTUM FIELD

Every “particle” has an underlying quantum field that reaches throughout all of space.

